PERFORMING RIGS

AutoPilot Firmware Validation Report

Document	AP-VVR-005
Revision	Rev A
Product	AutoPilot
Hardware Revision	Hardware Revision 2 (Batch-2 production receipt, PCB silk 52_P0_V01R04)
Firmware Revision	Autopilot_firm05-08 · 915 MHz · UI AutoPilot V01 R39 · Remote V01 R42
Validation Date	September 3, 2026
Prepared By	Nathan Destro, Performing Rigs
Location	Bench / shop

Purpose

Verify whether Noxon's proposed **Pilot Input Adapter polarity swap** (Phase 1; no resistor) resolves the **AP-VVR-004 Chain B false-trigger condition** without breaking intentional dry-contact fires.

This report does **not** reopen or revise AP-VVR-004 findings. It documents a new validation session.

Validation Summary

Validation Status	**OPEN**
Functional Result	**Major improvement — not a release pass**
Gate A — Chain B polarity retest	**OPEN — B-1 PASS; B-2 sustained idle PASS; B-2 connection transient FAIL (repeatable single unintended trigger)**

Disposition (this session): Corrective direction confirmed for Spark+ output (B-1). Validation remains **OPEN**. Batch-2 stays **quarantined**.

Release still blocked by:

1. Written PIA polarity standard + installed-base retrofit/replacement (F-005) 2. B-2 connection-induced unintended trigger (OPEN — F-002) 3. Original Pilot + swapped PIA **not tested** (if same adapter is used on Pilot) 4. Alfredo disposition on batch / field units

680 ohm: Deferred — not required for B-1 based on Phase 1.

Test configuration

Phase 1 — Polarity swap only (no resistor): Pilot Input Adapter leads reversed.

	Tip (TRS)	Sleeve	XLR note
Original (VVR-004 era)	Red pin 1	Black pin 2	As built
This session	Black pin 1 (GND/-)	Red pin 2 (+)	Per Iker 2026-09-02

Phase 2 — 680 ohm: Not performed — deferred; B-1 studio path resolved by polarity alone. Do not introduce resistor until connection transient is addressed or dismissed by Noxon.

Wiring paths (same as AP-VVR-004)

Test path	Phase 1 result	Phase 2
Chain A — Spark+ input: AP **|** Pilot Input Adapter **|** 25 4-Pin XLR **|** Relay Adapter **|** Spark+ input	**PASS** — 3300 mV, quiet (Spark+ **unpowered**, canonical). Powered Spark+ input continuous trigger (same with original PIA; unsupported topology — F-004)	N/A
Chain B-1 — Spark+ output: AP **|** Pilot Input Adapter **|** 25 4-Pin XLR **|** Spark+ output	**PASS** — 3300 mV, quiet ≥10 min; quiet with Spark+ powered; button fire immediate / expected	N/A
Chain B-2 — idle bench supply: AP **|** Pilot Input Adapter **|** 25 4-Pin XLR **|** Wire Lead **|** bench supply, not commanding	**Split:** sustained idle **PASS** (3300 mV); **connection transient FAIL** — first XLR connect activates AP once (repeatable). Commanded trip ≈ **10 V** (0/5/9 V no change; 10 V 0 mV + trigger)	N/A

Voltage observations

Condition	Phase 1 reading
Chain A idle INFO (mV) — Spark+ unpowered	**3300**
Chain B-1 idle INFO (mV) — polarity swapped	**3300**
Chain B-2 idle INFO (mV) — after settle	**3300**
Chain B-2 commanded 0 / 5 / 9 V	No INFO change; no trigger
Chain B-2 commanded **10 V**	**0 mV**; AP triggers
Intentional dry-contact	**10/10, 0 misses — path:** AP **|** PIA **|** 25 XLR **|** Wire Lead
Bench supply: output OFF (V at terminals)	**-0.125 V**
Bench supply: output ON @ 0 V	**0.04 V**
Bench supply: output ON @ 12 V	**12.08 V**
Bench supply: unpowered	**0.00 V**

Findings

F-001 — Chain B-1 resolved by Pilot Input Adapter polarity swap

With tip = black (pin 1) / sleeve = red (pin 2), **Spark+ output** idle no longer produces continuous false triggering. INFO stays ~3300 mV; intentional Spark+ button fire works. Supports Iker's directional-current explanation for Rev 2 relay-contact sensing. Addresses the primary studio hazard path from AP-VVR-004 Gate A (Chain B-1).

Severity: Info (mitigation validated Phase 1) **Status: PASS** (B-1)

F-002 — Chain B-2 connection-induced unintended trigger (OPEN)

Connecting XLR to Wire Lead with bench supply powered / ~0 V / not commanding causes **one** AP motor activation. **Repeatable** on first connect. After settle: INFO ~3300 mV, no sustained false triggers. Commanded voltage response coherent (trip ≈ **10 V**).

Do not label this a mere quirk. Requirement remains **zero unintended activation** when AutoPilot is armed. Sustained idle is acceptable; connection transient is a separate **FAIL**.

Sub-result	Verdict
Sustained idle after connect	**PASS**
Connection transient (first mate)	**FAIL** — OPEN

Severity: Medium (single unintended move on cable connect) **Status: OPEN**

F-003 — Intentional dry-contact still works after polarity swap

10 consecutive intentional fires via Wire Lead path, zero misses. Polarity swap did **not** fix powered paths by breaking relay-contact detection on Rev 2.

Severity: Info **Status: PASS**

F-004 — Powered Spark+ input AP (exploratory / unsupported)

AP Trigger · … · Relay Adapter · Spark+ input with Spark+ powered produces continuous triggering and Spark+ button LED on — same with original and swapped PIA wiring. Not a documented customer workflow. Possible backfeed into Spark+ input; not a release gate. Mispatch caveat: if XLR/adapters make powered-input vs output easy to confuse, document operator guidance later — do not divert engineering capital now.

Severity: Info (unsupported topology) **Status:** Documented

F-005 — Pilot Input Adapter polarity is now a product / installed-base issue

Original PIA build: tip/red pin 1, sleeve/black pin 2. Successful Phase 1 reverses that. Performing Rigs bench now shows swapped PIA works on **Rev 1 and Rev 2** for Spark+ output, dry-contact / Relay Contact, and 12/24 V (see matrix). Noxon must still confirm this as the **written required configuration** and say whether field adapters need retrofit or replacement.

Severity: Medium (process / field disposition) **Status: OPEN** — awaiting Noxon written polarity standard (bench compatibility **PASS** except Pilot)

F-006 — Rev 1 Relay Contact mode fires on Spark+ 24 V (observed)

With swapped PIA, Rev 1 in **Relay Contact** mode: dry-contact path works; operator also applied a **24 V Spark+ signal** while still in Relay Contact mode and AutoPilot **triggered**. Recorded as observed behavior (Rev 1 mode selector does not necessarily reject a 12/24 V source). Not treated as a Phase 1 fail.

Severity: Info **Status:** Documented

Compatibility matrix (before production PIA change)

Configuration	Status
Rev 2 + Spark+ output (swapped PIA)	**PASS**
Rev 2 + dry contact (swapped PIA)	**PASS** 10/10
Rev 2 + 12 V / 24 V commanded bench	**PASS**
Rev 1 + swapped PIA + Relay Contact mode	**PASS** (works). Note: 24 V from Spark+ while in Relay Contact also triggered AP (F-006)
Rev 1 + swapped PIA + 12/24 V mode	**PASS** — works and idles correctly
Rev 1 + swapped PIA + Spark+ output	**PASS** — works and idles correctly
Original Pilot + swapped PIA	**NOT TESTED** (if same adapter is used with original Pilot)

Questions for Noxon

Q-001 (priority — polarity standard)

Based on these results, should the **reversed** Pilot Input Adapter polarity (tip = GND/- / black / pin 1; sleeve = + / red / pin 2) now be considered **Noxon's required wiring configuration**? If so, confirm intended compatibility with **Rev 1 AutoPilot, Rev 2 AutoPilot, Pilot, relay-contact sources, and 12/24 V powered sources**. Performing Rigs already has Pilot Input Adapters in customer use — is this a production change, and do existing adapters require retrofit or replacement?

Q-002

Chain B-2: first XLR connection to an idle ~0 V bench supply fires AutoPilot **once** (armed). Is that expected? Recommended connect / power-up sequence?

Q-003

Under polarity-swapped PIA, commanded bench voltage tripped at ~**10 V** (INFO 0 mV). Rev 2 explicit **12 V / 24 V bench** fires **PASS**. Confirm intended trip band for this polarity.

Q-004 (carry forward)

Disposition for Rev 1 sold units, Rev 1 inventory, and Revision 2 Batch-2 (Alfredo).

Q-005

680 ohm cable — still recommended for any interface, or optional now that B-1 is quiet with polarity alone?

Engineering Assessment

Dimension	Conclusion
Chain B-1 (Spark+ output) polarity retest	**PASS**
Chain B-2 sustained idle	**PASS**
Chain B-2 connection transient	**FAIL** — OPEN (F-002)
Chain A (Spark+ input, unpowered)	**PASS** (3300 mV quiet)
Intentional dry-contact	**PASS** (10/10 Wire Lead)
680 ohm Phase 2	**Deferred**
PIA polarity as product standard	**OPEN** (F-005) — Noxon written confirm
Rev 1 + swapped PIA (Relay Contact / 12–24 V / Spark+ out)	**PASS**
Original Pilot + swapped PIA	**NOT TESTED**
Hardware Revision 2 Batch-2 release	**NOT CLEARED**
Overall Validation Status	**OPEN**
Overall Functional Result	Major improvement / corrective direction confirmed — **not** released

Shipment posture: Do **not** sell from Batch-2. Swapped PIA looks like a **universal AutoPilot correction** on Rev 1 and Rev 2 for the tested sources — not Rev-2-only. Still need Noxon's written polarity standard (and field-adapter policy), F-002 connect transient, Alfredo, and Pilot if that adapter is shared.

Revision History

Rev	Date	Description
A	2026-09-03	Issued — Phase 1 polarity retest; B-1 PASS; B-2 connection transient OPEN FAIL; Rev 1/Rev 2 compatibility matrix PASS (Pilot untested); 680 ohm deferred